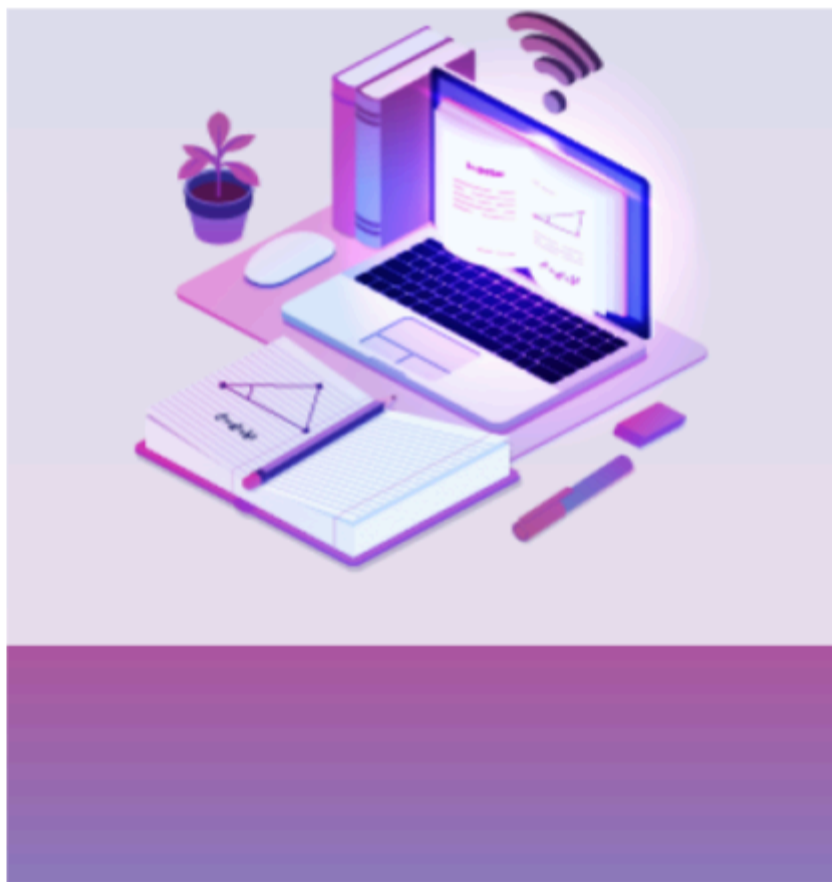


CAT Time, Speed, Distance & Work Formulas PDF

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CAT Time, Speed, Distance & Work Formulas

- Time, Distance and Work is the most important topic for CAT Quant Section & all competitive exams.
- The questions from this topic vary from easy to difficult.
- This formula sheet covers the most importance tips that helps you to answer the questions in a easy, fast and accurate way

$$Distance = Speed \times Time$$

$$Speed = \frac{Distance}{Time} \quad \Bigg| \quad Time = \frac{Distance}{Speed}$$

- While converting the speed in m/s to km/hr, multiply it by $\left(\frac{18}{5}\right) \Rightarrow 1 \text{ m/s} = 3.6 \text{ km/h}$

- While converting km/hr into m/sec, we multiply by

$$\left(\frac{5}{18} \right)$$

- If the ratio of the speeds of A and B is $a : b$, then
 - The ratio of the times taken to cover the same distance is $1/a : 1/b$ or $b : a$.
 - The ratio of distance travelled in equal time intervals is $a : b$,

$$\text{Average Speed} = \frac{\text{Total Distance Travelled}}{\text{Total Time Taken}}$$

- If a part of a journey is travelled at speed S_1 km/hr in T_1 hours and remaining part at speed S_2 km/hr in T_2 hours then,

$$\text{Total distance travelled} = S_1 T_1 + S_2 T_2 \text{ km}$$

$$\text{Average Speed} = \frac{S_1 T_1 + S_2 T_2}{T_1 + T_2} \text{ km/hr}$$



- If D_1 km is travelled at speed S_1 km/hr, and D_2 km is travelled at speed S_2 km/hr then,

$$\text{Average Speed} = \frac{D_1 + D_2}{\frac{D_1}{S_1} + \frac{D_2}{S_2}} \text{ km/hr}$$

- In a journey travelled with different speeds, if the distance covered in each stage is constant, the average speed is the harmonic mean of the different speeds.
- Suppose a man covers a certain distance at x km/hr and an equal distance at y km/hr. Then the average speed during the whole journey is $\frac{2xy}{x+y}$ km/hr
- In a journey travelled with different speeds, if the time travelled in each stage is constant, the average speed is the arithmetic mean of the different speeds.

- If a man travelled for a certain distance at x km/hr and for equal amount of time at the speed of y km/hr then the average speed during the whole journey is

$$\frac{x+y}{2} \text{ km/hr}$$

Constant Distance

Let the distance travelled in each part of the journey be d_1, d_2 & d_3 and so on till d_n and the speeds in each part be s_1, s_2, s_3 and so on till s_n

If $d_1 = d_2 = d_3 = \dots = d_n = d$, then the average speed is the harmonic mean of the speeds s_1, s_2, s_3 and so on till s_n .



Constant Time

Let the distance travelled in each part of the journey be d_1, d_2 and d_3 and so on till d_n and the time taken for each part be t_1, t_2, t_3 and so on till t_n .

If $t_1 = t_2 = t_3 = \dots = t_n = t$, then the average speed is the arithmetic mean of the speeds s_1, s_2, s_3 and so on till s_n .

Clocks

→ Calculating the angle/position of the hands

- Speed of hour hand = 0.5° per minute
- Speed of minute hand = 6° per minute
- Relative speed of two hands = 5.5° per minute
- The angle (in degrees) between hour hand and minute hand at time H: M can be represented

$$\text{as: } \theta = \left| \frac{11}{2}M - 30H \right|^\circ$$

→ In a 12-hour period:

- The hour hand and the minute hand meet 11 times
- A 180^0 angle is formed between the two hands 11 times
- A 90^0 angle is formed between the two hands 22 times

→ In a well-functioning clock, both hands meet

after every $\frac{720}{11}$ Mins.

→ It is because the relative speed of the minute hand

with respect to the hour hand $= \frac{11}{2}$ degrees per

minute.



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Erroneous Clocks

→ An erroneous clock is a clock which loses or gains time at a constant rate.

→ In case of an erroneous clock losing/gaining 'x' sec per minute,

- It will lose/gain 'x' minutes per hour.
- It will show the correct time after every '720/x' hours.
- The clock will show the same time again after 'y' hours where:

$$y = \frac{720}{60+x} \text{ if the clock gains time.}$$

$$y = \frac{720}{60-x} \text{ if the clock loses time.}$$

- If the clock is set right at 'Q' AM/PM. Then the time 'T' shown by the clock after 'h' hours pass on a correct clock would be:

$$T = Q + (h \times x) \text{ if the clock gains time.}$$

$$T = Q - (h \times x) \text{ if the clock loses time.}'$$

- If the clock is set right at 'Q' AM/PM. If 'h' hours pass on the erroneous clock, then the actual time 'T' shown by a correct clock would be:

$$T = Q + y ; y(\text{in hours})$$

$$= \frac{60h}{60+x} \text{ if the clock gains time}$$

$$= \frac{60h}{60-x} \text{ if the clock loses time.}$$



Circular Tracks

If two people are running on a circular track with speeds in ratio $a:b$ where a and b are co-prime, then

→ They will meet at $a + b$ distinct points if they are running in opposite directions.

→ They will meet at $|a - b|$ distinct points if they are running in same direction

- If two people are running on a circular track having perimeter 'l', with speeds 'm' and 'n',

$$\text{The time for their first meeting} = \frac{l}{(m+n)}$$

(when they are running in opposite directions)

$$\text{The time for their first meeting} = \frac{l}{|(m-n)|}$$

(when they are running in the same direction)

- If a person P starts from A and heads towards B and another person Q starts from B and heads towards A and they meet after a time 't' then,

$$t = \sqrt{(x \times y)}$$

where x = time taken (after meeting) by P to reach B
and y = time taken (after meeting) by Q to reach A.

- A and B started at a time towards each other. After crossing each other, they took $T_1 \text{ hrs}$, $T_2 \text{ hrs}$ respectively to reach their destinations. If they travel at constant speed S_1 and S_2 respectively all

over the journey, then
$$\frac{S_1}{S_2} = \sqrt{\frac{T_2}{T_1}}$$



Trains

Two trains of length L_1 and L_2 travelling at speeds S_1 and S_2 cross each other in a time

$$= \frac{L_1 + L_2}{S_1 + S_2} \text{ (If they are going in opposite directions)}$$

$$= \frac{L_1 + L_2}{|S_1 - S_2|} \text{ (If they are going in the same directions)}$$

Time & Work

$\Rightarrow$ If X can do a work in 'n' days, the fraction of work X does in a day is $\frac{1}{n}$

$\Rightarrow$ If X can do work in 'x' days, and Y can do work in 'y' days, then the number of days taken by both of them together is $\frac{x \times y}{x + y}$

⇒ If M_1 men work for H_1 hours per day and worked for D_1 days and completed W_1 work, and if M_2 men work for H_2 hours per day and worked for D_2 days and completed W_2 work, then

$$\frac{M_1 H_1 D_1}{W_1} = \frac{M_2 H_2 D_2}{W_2}$$

Boats & Streams

⇒ If the speed of water is 'W' and speed of a boat in still water is 'B'

⇒ Speed of the boat (downstream) is $B+W$

⇒ Speed of the boat (upstream) is $B-W$

The direction along the stream is called **downstream**.

And, the direction against the stream is called **upstream**.



⇒ If the speed of the boat downstream is x km/hr and the speed of the boat upstream is y km/hr, then

⇒ Speed of the boat in still water = $\frac{x + y}{2}$ km/hr

⇒ Rate of stream = $\frac{x - y}{2}$ km/hr

⇒ While converting the speed in m/s to km/hr,

multiply it by $\left(\frac{18}{5}\right) \Rightarrow 1 \text{ m/s} = 3.6 \text{ km/h}$

⇒ While converting km/hr into m/sec,

we multiply by $\left(\frac{5}{18}\right)$

Pipes & Cisterns

⇒ Inlet Pipe : A pipe which is used to fill the tank is known as Inlet Pipe.

⇒ Outlet Pipe : A pipe which can empty the tank is known as outlet pipe.

- If a pipe can fill a tank in 'x' hours then the part filled per hour = $\frac{1}{x}$
- If a pipe can empty a tank in 'y' hours, then the part emptied per hour = $\frac{1}{y}$
- If a pipe A can fill a tank in 'x' hours and pipe can empty a tank in 'y' hours, if they are both active at the same time, then

$$\text{The part filled per hour} = \frac{1}{x} - \frac{1}{y} \text{ (If } y > x \text{)}$$

$$\text{The part emptied per hour} = \frac{1}{y} - \frac{1}{x} \text{ (If } x > y \text{)}$$

Some Tips and Tricks

- Some of the questions may consume a lot of time.
While solving, write down the equations without any



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errors once you fully understand the given problem. The few extra seconds can help you avoid silly mistakes.

- Check if the units of distance, speed and time match up. If you see yourself adding a unit of distance like m to a unit of speed m/s, you would realise you have possibly missed a term.
 - Choose to apply the concept of relative speed wherever possible since it can greatly reduce the complexity of the problem.
 - In time and work, while working with equations, ensure that you convert all terms to consistent units like man-hours.
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